

DFIR Memory-Triage Report — contact_me (1 GiB raw RAM)

CONFIDENTIAL — Internal DFIR / Examiner use · Agentropix-SIFT sealed report

Memory-forensics triage of a 1 GiB raw RAM capture. Unprofileable image:
Volatility3 2.28.0 could not validate a Windows kernel symbol table, so process / network / injection / service enumeration returned placeholder or empty results.
No clean-or-compromised determination is possible from this capture.

Field	Value
Report ID	e9763e7eda4892b0895631ebd24b915373ec31dbc85e10dffb1d1ed8566a10908
Case ID	CTF-CONTACT-ME-MEM
Case name	CTF contact_me (raw memory)
Examiner	victor.galvan
Profile	full
Incident type	dfir
Severity	Medium
Snapshot at (UTC)	2026-06-07T12:40:26Z
Generated at (UTC)	2026-06-07T12:40:26Z
MCP host	<TAILNET-HOST>
Seal	hmac-sha256:caa3c5618997c893599d6b5fddea003ea9cc0d12c5a2a48c216920264629f779

Audiences served: CISO / stakeholder · SOC / blue team · Red team · Audit.

1. Executive Summary

A 1 GiB raw memory capture (`/cases/contact_me/contact_me`) was registered, hashed, and triaged through the Agentropix-SIFT memory pipeline. The image is unprofileable: Volatility3 2.28.0 could not auto-detect or validate a Windows kernel symbol table for this capture. Every kernel-dependent plugin (pslist, netscan, malfind, svcsan, cmdline) therefore returned placeholder rows, empty result sets, or a requirements error rather than real artefacts. This is an honest negative-control outcome — not a "clean" verdict. No processes, sockets, injected regions, services, IOCs, or timeline events could be resolved. One medium-severity finding records the unprofileable condition itself; it was indexed and (in a simulated examiner sign-off, demo only) sealed into this report.

Key finding — The capture cannot be analysed with the current Volatility3 symbol set; remediation is to re-acquire with a known OS/build or supply a matching symbol table, not to draw conclusions about compromise.

2. KPI Summary

KPI	Value	Detail
Approved findings	1	F-CONTACTME-001 (🟡 medium) — records the unprofileable outcome
Hosts in scope	1	contact_me (single RAM capture, OS unresolved)
IOCs catalogued	0	none resolvable — no validated kernel layer to scan
Attacker dwell	N/A	no timeline resolvable from this capture
MITRE techniques	0	none observed — kernel-dependent plugins did not return artefacts
Initial access	Inconclusive	not determinable from an unprofileable RAM snapshot

Host roster: `contact_me` — 1 GiB raw RAM, OS/build unidentified by Volatility3 2.28.0.

3. Risk Matrix

No exploitable risk could be scored: with no validated kernel symbol table, no process, network, injection, service, IOC, or timeline evidence was recovered. The only scored item is the analytic risk that the capture yields no determination.

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 Severe	.	.	.	.	.
4 Major	.	.	.	.	.
3 Moderate	.	.	F1 	.	.
2 Minor	.	.	.	.	.
1 Negligible	.	.	.	.	.

Scored findings

Ref	Risk	Impact	Likelihood	Score	Severity
F1	Capture unprofileable — no clean/compromise determination possible	3	3	9	 Medium

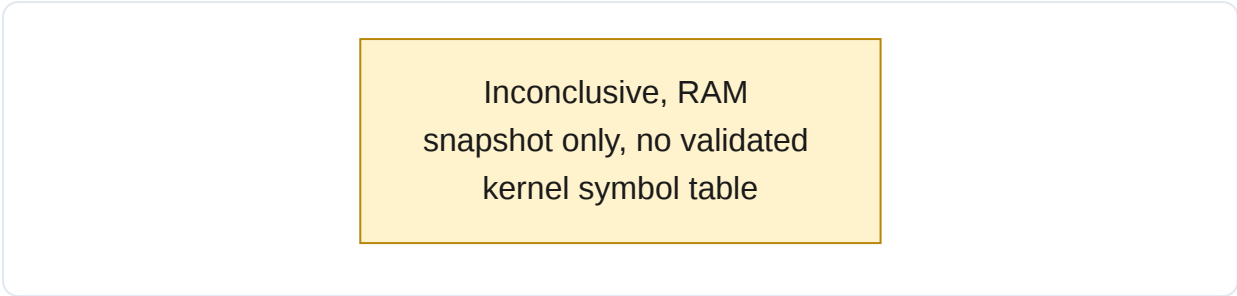
4. Key Findings

-  F-CONTACTME-001 — Memory image unprofileable (Medium) — Volatility3 2.28.0 could not validate a Windows kernel symbol table, so pslist / netscan / malfind / svcsan returned placeholder or empty results and no clean-or-compromised determination is possible.

5. Attack Chain & MITRE ATT&CK

No attack chain could be reconstructed. Kernel-dependent enumeration did not return resolvable artefacts, so there is no execution, persistence, or C2 evidence to chain. The diagram below honestly marks the inconclusive state.

5.1 Attack chain



5.2 MITRE ATT&CK techniques

Tactic	Technique ID	Technique	Evidence / how observed
—	—	None observed	RAM snapshot unprofileable — no kernel layer to map techniques against

6. IOC Catalogue

No indicators of compromise could be extracted. With no validated symbol table, windows.netscan , windows.malfind , and windows.svcscan had no kernel layer to scan and returned empty result sets; the cmdline plugin emitted a non-JSON requirements error.

Type	Value	Role	Confidence	MITRE	Source
—	—	—	—	—	No IOCs resolvable — unprofileable capture (netscan/malfind/svcscan empty, cmdline error)

Negative space (surveyed, not observed): network sockets (netscan: socket_count 0), injected/RWX regions (malfind: hit_count 0), services (svcsan: service_count 0), command lines (cmdline: non-JSON error). All four are gated by the missing kernel symbol table, so none is a confirmed "absent" — only "not resolvable".

6.1 IOC provenance

No IOC provenance, no validated kernel layer to scan

7. Host Artefacts

A single host, `contact_me`, is represented by one 1 GiB raw RAM capture. Volatility3 2.28.0 could not validate the kernel layer / symbol table for this image, so the process list returned 11 placeholder rows (all `pid:0`, `name:"unknown"`), and the service scan returned zero rows. malfind reported `hit_count 0` for the same reason — there were no resolvable RWX/injected VAD regions because there was no kernel layer to enumerate. None of these are clean-result signals.

7.1 Host — contact_me

Processes (pslist)

PID	PPID	Image	Started (UTC)	State	Wow64	Notes	Source
0	0	unknown	—	—	—	placeholder row, kernel symbol table not validated (11 such rows)	get_pslis t

Services (svcsan)

Service	Image path	Install (UTC)	Class	Source
—	—	—	—	no service list captured (svcsan service_count 0, requirements error)

malfind: hit_count 0 — no injected / RWX VAD regions resolvable (no validated kernel layer); not a confirmed-clean result.

7.2 Process Tree

unknown PID 0, single empty root, no profile match

build_process_tree ran (process_count 11, root_count 1, orphan_count 0, suspicious_count 0) but was fed the placeholder rows, so the tree collapses to one empty pid:0 root with no children and no LOLBin / suspicious-parent flags.

8. Network Artefacts

No network artefacts could be recovered. windows.netscan returned socket_count 0 with a requirements error — there was no validated kernel layer for it to scan. This is not a "no connections" finding.

Process (PID)	Local	Remote	State	Purpose	Source
—	—	—	—	—	No sockets resolvable — netscan socket_count 0, kernel symbol table not validated

netscan raw_stderr: Unable to validate the plugin requirements: [plugins.NetScan.kernel.layer_name, plugins.NetScan.kernel.symbol_table_name] .

9. Detailed Findings

9.1 F-CONTACTME-001 — Memory image unprofileable (Volatility3 symbol table not validated)

Severity	 Medium (confidence 0.90)
Host	contact_me
Technique	None mapped (no kernel layer to attribute to)
Status	Indexed; sealed into report (examiner sign-off SIMULATED — demo only)

Volatility3 2.28.0 could not auto-detect or validate a Windows kernel symbol table for the `contact_me` raw capture. As a result every kernel-dependent plugin failed to produce real artefacts: `get_pslist` returned 11 placeholder rows (`pid:0` , `name:"unknown"`); `get_netscan` returned `socket_count 0` ; `get_malfind` returned `hit_count 0` ; `get_svcscan` returned `service_count 0` ; and `run_volatility plugin=cmdline` produced a non-JSON requirements error. `build_process_tree` correlated the same placeholder rows into a single empty root. No clean-or-compromised determination is possible. The correct interpretation is a tooling/profile mismatch (re-acquire with a known OS/build or supply a matching symbol table), not a statement about the host's security state.

Evidence — evidence SHA-256

1ab5eb6c3b87a0604f75a00cb4a64d91aaf7ab4e303bb7337b40f7e3df8ad61a (/cases/contact_me/contact_me , 1073741824 bytes = 1 GiB) · sha256

1ab5eb6c3b87a0604f75a00cb4a64d91aaf7ab4e303bb7337b40f7e3df8ad61a · captures: get_pslist / get_netscan / get_malfind / get_svcscan / build_process_tree / run_volatility(cmdline). Finding HMAC seal hmac-

sha256:caa3c5618997c893599d6b5fddea003ea9cc0d12c5a2a48c216920264629f779 .

10. Timeline

No timeline could be reconstructed. The sealed report's timeline section is empty (count 0) because no events could be resolved from an unprofileable capture.

Time (UTC)	Host	Event	Technique	Source
—	—	No timeline resolvable — unprofileable RAM capture (sealed timeline count 0)	—	—

11. Agentropix Performance

The pipeline executed end-to-end (doctor → health → case_init → case_activate → evidence_register → 6 memory captures → record_finding → simulated approval → report_generate). The tooling worked; the capture itself was the limiting factor. The MCP server answered HTTP 200 with a live `tool_count` of 72 (canonical is 71; the live count can read 72).

Metric	Value	Detail
Run time	Not separately captured	per-stage timings not recorded in this run
MCP tool calls	13+	doctor/health + init/activate/status + evidence + 6 captures + finding + approval + report
Evidence size	1073741824 bytes	1 GiB raw RAM (/cases/contact_me/contact_me)
Disk recall	72/72 (100%)	canonical (.crew/facts.md)
Memory recall	108/118 (91.5%)	canonical (.crew/facts.md)
Test suite	4464	canonical (.crew/facts.md)

Per-stage timing

Stage	Capture	Duration	Result
—	—	—	no per-stage timings captured

12. Coverage Attestation

Disk and memory recall figures are canonical regression metrics for the Agentropix-SIFT suite, not metrics of this single unprofileable run. The evidence gate was exercised: the real finding index was gated by a one-shot `index_findings` evidence-gate token.

Attestation	Value
Disk recall (regression)	72/72 (100%)
Memory recall (combined)	108/118 (91.5%)
Test suite	4464
Evidence gate	Enforced — one-shot <code>index_findings</code> token required for the real index

This report honestly records an unprofileable capture. The single approved finding documents that no determination was possible; it does not assert the host is clean or compromised. Volatility3 2.28.0 could not validate a kernel symbol table, which is the controlling fact for every section above.

13. Recommendations

1. [P1] Re-acquire or re-identify the OS/build of the `contact_me` host and supply Volatility3 with a matching kernel symbol table (ISF), then re-run `pslist/netscan/malfind/svcscan`; treat the current empty/placeholder outputs as not-resolvable, never as clean.
2. [P2] Before drawing any conclusion from a memory image, confirm a populated process list — not an HTTP 200 — as the true signal that a kernel profile matched; gate downstream findings on `process_count` with non-placeholder rows.

3. [P3] Preserve the registered evidence (SHA-256 `1ab5eb6c...`) and chain of custody so a re-run with a correct symbol table can be compared byte-for-byte against this baseline.

► Appendix — methodology, tool versions, chain of custody